

exp4.py - E:/machine_learning_lab_exp/exp4.py (3.10.11)

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```
import numpy as np

# Input and Output
X = np.array([[0,0],[0,1],[1,0],[1,1]])
y = np.array([[0],[1],[1],[0]])

# Initialize weights
w1 = np.random.rand(2,2)
w2 = np.random.rand(2,1)

# Training
for i in range(5000):
    h = 1/(1+np.exp(-np.dot(X, w1)))
    o = 1/(1+np.exp(-np.dot(h, w2)))
    w2 += np.dot(h.T, (y-o))
    w1 += np.dot(X.T, np.dot((y-o), w2.T))

print("Final Output:")
print(o.round())
```

IDLE Shell 3.10.11

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```
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr 5 2023, 00:38:17)
[MSC v.1929 64 bit (AMD64)] on win32
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nformation.
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===== RESTART: E:/machine_learning_lab_exp/exp4.py =====
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Final Output:
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[1.]
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